

8. (a) When a mixture of iron(III) oxide and aluminium powder (Thermit mixture) is heated, there is a violent reaction. The reaction is carried out in a tube surrounded by a mound of sand because the temperature reaches 2500°C . A bead of iron is recovered from the sand. The picture below shows the reaction taking place in a darkened room.



- (i) Give the reason why the iron formed in the reaction is molten. [1]

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.....

- (ii) Complete and balance the **symbol** equation for this reaction. [2]



- (iii) State which of the substances is oxidised. Give the reason for your choice. [1]

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- (iv) When a mixture of magnesium oxide and aluminium powder is heated, there is no reaction.

List iron, magnesium and aluminium in order of reactivity. [1]

Most reactive

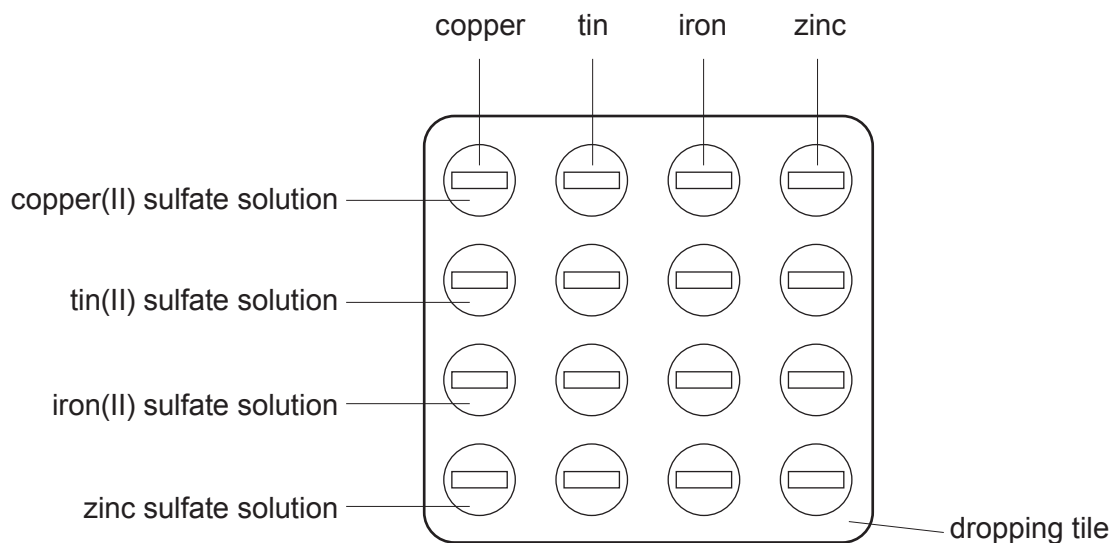
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Least reactive



- (b) Some metals are more reactive than others. A more reactive metal displaces a less reactive metal from its compounds.

A student was given tin, iron, copper and zinc and solutions of the metal sulfates. Using a dropping pipette, she put a little of one of the sulfate solutions in four of the depressions of the dropping tile. She did this for each solution in turn. She then put a piece of metal foil in each of the solutions, as shown below.



- (i) Put a tick (✓) next to the question which **best** describes the investigation the student is carrying out. [1]

Which displacement is the most exothermic?

☐

Which metal can displace copper from solution?

☐

What is meant by the reactivity series?

☐

What are the positions of the four metals in the reactivity series?

☐


- (ii) The student recorded the results by putting a tick (✓) next to a mixture which showed signs of a reaction and a cross (X) next to a mixture which showed no signs of a reaction.

The student concluded that:

tin displaces copper
iron displaces tin
iron displaces copper
zinc displaces iron

Give the **letter** of the tile below which shows the results she recorded.

[1]

Letter

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

A

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

B

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

C

	copper	tin	iron	zinc
copper(II) sulfate solution	☒	☒	☒	☒
tin(II) sulfate solution	☒	☒	☒	☒
iron(II) sulfate solution	☒	☒	☒	☒
zinc sulfate solution	☒	☒	☒	☒

D

- (iii) Another student said that not all of the tests were necessary. Give **one** example of a test not needed. Explain your choice.

[2]

Example

Explanation

.....



Examiner
only

(c) Copper displaces silver from a solution of silver nitrate, AgNO_3 , to form copper(II) nitrate solution.

(i) Describe **one** change the student would **see** during this displacement reaction. [1]

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.....

(ii) Write a balanced **symbol** equation for this reaction. [2]

.....

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0

1 H Hydrogen 1										4 He Helium 2																									
7 Li Lithium 3		9 Be Beryllium 4		11 B Boron 5		12 C Carbon 6		14 N Nitrogen 7		16 O Oxygen 8		19 F Fluorine 9		20 Ne Neon 10																					
23 Na Sodium 11		24 Mg Magnesium 12		27 Al Aluminium 13		28 Si Silicon 14		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17		40 Ar Argon 18																					
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		127 I Iodine 53		131 Xe Xenon 54			
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																															
																			Key																

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

